

KAKTOVIK, AK, USA

WMO: 700860

Lat: 70.134N

Lon: 143.577W

Elev: 1

StdP: 101.32

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27401

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-37.7	-35.6	-42.1	0.1	-37.1	-39.6	0.1	-35.1	21.7	-19.2	19.8	-18.0	2.6	260	1.083

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	15.5	12.9	12.7	10.9	10.6	9.3	13.2	15.1	11.1	12.5	9.4	10.5	4.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	8.7	14.8	10.0	7.6	12.0	8.5	6.9	10.2	37.0	15.3	31.8	12.5	27.8	10.4	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.2	15.5	13.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-37.1	16.5	3.1	5.7	-39.3	20.6	-41.2	23.9	-42.9	27.1	-45.2	31.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.9	-22.4	-23.3	-22.2	-13.4	-3.7	1.7	6.5	5.7	1.8	-4.7	-14.3	-19.1		
	DBStd	12.38	7.36	7.51	8.21	7.38	4.49	2.81	3.00	3.96	2.72	4.33	6.15	7.15		
	HDD10.0	6909	1003	934	999	705	423	249	116	146	245	457	728	903		
	HDD18.3	9931	1262	1167	1257	953	682	499	367	394	495	716	978	1161		
	CDD10.0	22	0	0	0	3	0	0	7	12	0	0	0	1		
	CDD18.3	3	0	0	0	1	0	0	0	3	0	0	0	0		
	CDH23.3	45	0	0	0	5	0	0	0	36	0	0	0	4		
	CDH26.7	14	0	0	0	2	0	0	0	10	0	0	0	2		
(14) Wind	WSAvg	5.6	6.0	5.7	5.1	5.0	5.1	5.1	4.8	5.1	5.7	6.7	6.5	6.2		
(15) Precipitation	PrecAvg	115	6	3	4	3	5	10	21	22	14	14	8	5		
	PrecMax	229	45	19	37	9	30	33	76	57	44	36	29	17		
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	53	9	4	7	2	7	9	20	16	11	11	7	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	-1.0	7.7	20.9	5.0	12.9	17.8	28.0	11.7	6.3	0.6	8.6		
		MCWB	1.9	-1.2	5.7	17.7	3.3	10.0	13.7	26.9	8.7	4.5	-0.3	7.5		
	2%	DB	-5.2	-5.8	-0.1	3.3	3.2	9.2	14.8	15.0	9.0	2.6	-2.4	-2.5		
		MCWB	-5.6	-6.3	-1.1	1.1	2.1	7.5	12.4	12.7	7.1	1.8	-3.1	-3.5		
	5%	DB	-10.1	-9.9	-6.7	-1.5	2.3	6.9	12.5	12.4	6.9	1.1	-4.1	-7.2		
		MCWB	-10.6	-10.4	-7.1	-2.4	1.4	5.6	11.1	10.8	5.7	0.6	-4.7	-7.8		
	10%	DB	-12.7	-13.1	-11.0	-4.1	1.4	5.0	10.6	10.5	5.3	0.1	-5.9	-9.7		
		MCWB	-13.2	-13.7	-11.7	-4.8	0.6	4.1	9.5	9.3	4.5	-0.3	-6.3	-10.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	-1.2	6.0	17.5	3.6	10.0	14.4	27.0	9.1	4.7	-0.3	7.0		
		MCDB	4.0	-0.7	7.6	19.7	4.5	12.7	16.7	28.1	11.5	6.4	0.5	8.7		
	2%	WB	-5.8	-6.1	-1.4	1.6	2.3	7.7	12.6	13.0	7.2	1.9	-3.1	-3.4		
		MCDB	-5.3	-5.8	0.1	3.3	3.1	9.1	14.5	14.7	8.7	2.4	-2.6	-2.4		
	5%	WB	-10.6	-10.2	-7.1	-2.3	1.5	5.7	11.1	11.0	5.8	0.6	-4.7	-7.7		
		MCDB	-10.2	-9.7	-6.7	-1.7	2.1	6.8	12.6	12.2	6.8	0.9	-4.1	-7.1		
	10%	WB	-13.1	-13.5	-11.6	-4.8	0.8	4.2	9.6	9.3	4.6	-0.3	-6.2	-10.1		
		MCDB	-12.7	-12.9	-11.0	-4.1	1.3	4.9	10.6	10.3	5.3	0.1	-5.8	-9.6		
(35) Mean Daily Temperature Range	MDBR		6.6	6.4	8.0	7.9	4.7	3.7	5.0	4.0	3.0	4.2	5.7	6.1		
	5% DB	MCDBR	12.3	10.5	11.8	11.3	5.1	7.3	9.3	7.9	6.3	4.2	7.2	9.3		
		MCWBR	11.7	10.3	11.5	10.1	4.3	5.7	7.2	6.0	4.7	3.7	6.9	9.0		
	5% WB	MCDBR	12.2	10.6	11.7	10.9	5.0	6.9	9.0	7.4	5.9	4.0	7.0	9.1		
		MCWBR	11.6	10.4	11.4	9.9	4.2	5.5	7.2	5.8	4.6	3.6	6.7	8.8		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.202	0.244	0.301	0.327	0.348	0.357	0.339	0.304	0.221	N/A	N/A		
	taud		N/A	2.051	2.180	2.087	2.115	2.234	2.380	2.480	2.495	2.179	N/A	N/A		
	Ebn at Noon		0	544	777	820	843	837	818	793	721	507	0	0		
	Edh at Noon		0	38	71	111	123	114	96	77	56	32	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.37	2.00	4.29	5.95	6.49	5.17	3.08	1.64	0.48	0.00	0.00		
	RadStd		0.00	0.03	0.10	0.24	0.46	0.50	0.43	0.30	0.14	0.06	0.00	0.00		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	-335	-348	N/A	N/A	N/A

Nomenclature: See separate page